Homework 1

1. Write regular expressions for the following languages WITHOUT USING SHORTCUTS (i.e. no \d, \b, etc) (2.6.1 - 2.6.5)

    a. the set of all alphabetic strings
    b. the set of all lower case alphabetic strings ending in a `b`
        a. Example matches: brb, hijab, b
        b. Example non-matches: hello, ba
    c. the set of all strings from the alphabet *`a, b`* such that each ***a*** is immediately preceded by and immediately followed by a `b`
        a. Example matches: bbbbabbb, babab, bbbbbabbbabbbabbab
        b. Example non-matches: baab, aba
2. Write regular expressions for the following languages. By "word", we mean an alphabetic string separated from other words by whitespace, any relevant punctuation, line breaks, and so forth. (2.6.6 - 2.6.9)

    a. the set of all strings with two consecutive repeated words
        a. Example matches: Humbert Humbert, the the
        b. Example non-matches: big bug, the big bug
    b. all strings that start at the beginning of the line with an integer and that end at the end of the line with a word
        a. Example matches: 5 times I went to the Seven-Eleven
        b. Example non-matches: 5 times I went to the 7-11
    c. all strings that have both the word grotto and the word raven in them (but not, e.g., words like grottos that merely contain the word grotto)
        a. Example matches: raven grotto, The raven flew over the grotto, grotto raven
        b. Example non-matches: Quoth the raven grottos are cool!, raven's grotto
    d. write a pattern that places the first word of an English sentence in a register. Deal with punctuation.

3. Edit Distance (2.9.1)
    a. Compute the edit distance (using insertion cost 1, deletion cost 1, substitution cost 2) of "`leda`" to "`deal`". Show your work (using the edit distance grid).
    b. Figure out whether `drive` is closer to `brief` or to `divers` and what the edit distance is to each.
    c. Now implement a minimum edit distance algorithm in Java and use your hand-computed results to check your code.